

DRY-RUN GUIDE | 03_gcd.java
Source: java\2nd_belt\03_gcd.java

CORE BEHAVIOR

The static gcd method performs Euclid's remainder algorithm and returns a. main reads two integers and prints that result. Inputs are not normalized to absolute values.

REPRESENTATIVE INPUT / CALL

Input: '48 18'.

STEP TRACE

(48,18)->(18,12)->(12,6)->(6,0); return a=6.

EXPECTED OUTPUT / RETURN

'6'

ACTUAL SOURCE EXCERPT

Verbatim lines from the source; long lines wrap. Source files were not modified.

```
001 | /*
002 | Question:
003 | Write a program to find the greatest common divisor (GCD) of two integers.
004 |
005 | Input:
006 | Two integers A and B.
007 |
008 | Output:
009 | Print the GCD of A and B.
010 | */
011 |
012 | import java.util.Scanner;
013 |
014 | public class GCD {
015 |     public static int gcd(int a, int b) {
016 |         while (b != 0) {
017 |             int temp = a % b;
018 |             a = b;
019 |             b = temp;
020 |         }
021 |         return a;
022 |     }
023 |
024 |     public static void main(String[] args) {
025 |         Scanner sc = new Scanner(System.in);
026 |         int a = sc.nextInt();
027 |         int b = sc.nextInt();
028 |
029 |         System.out.println(gcd(a, b));
030 |         sc.close();
031 |     }
032 | }
033 |
```